

Scaling for Cultural Equity: Investigating the Impact of Training Data Size on Cultural Bias in Vision-Language Models

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Introduction

Problem:

- Vision-language models (VLMs) are deployed in global contexts, but often exhibit **strong performance biases favoring Western cultural contexts**, and notably weaker results in underrepresented regions (e.g.: Africa and Asia).^{[1][2]}
- These biases stem from imbalanced training data, reinforcing dominant cultural biases, while improving performance.^[3]
- Bias is rarely studied from the perspective of scaling laws^[3], i.e. increasing fine-tuning data size amplifies existing cultural biases.

Goal: Investigate whether more training data increases the model's cultural bias.

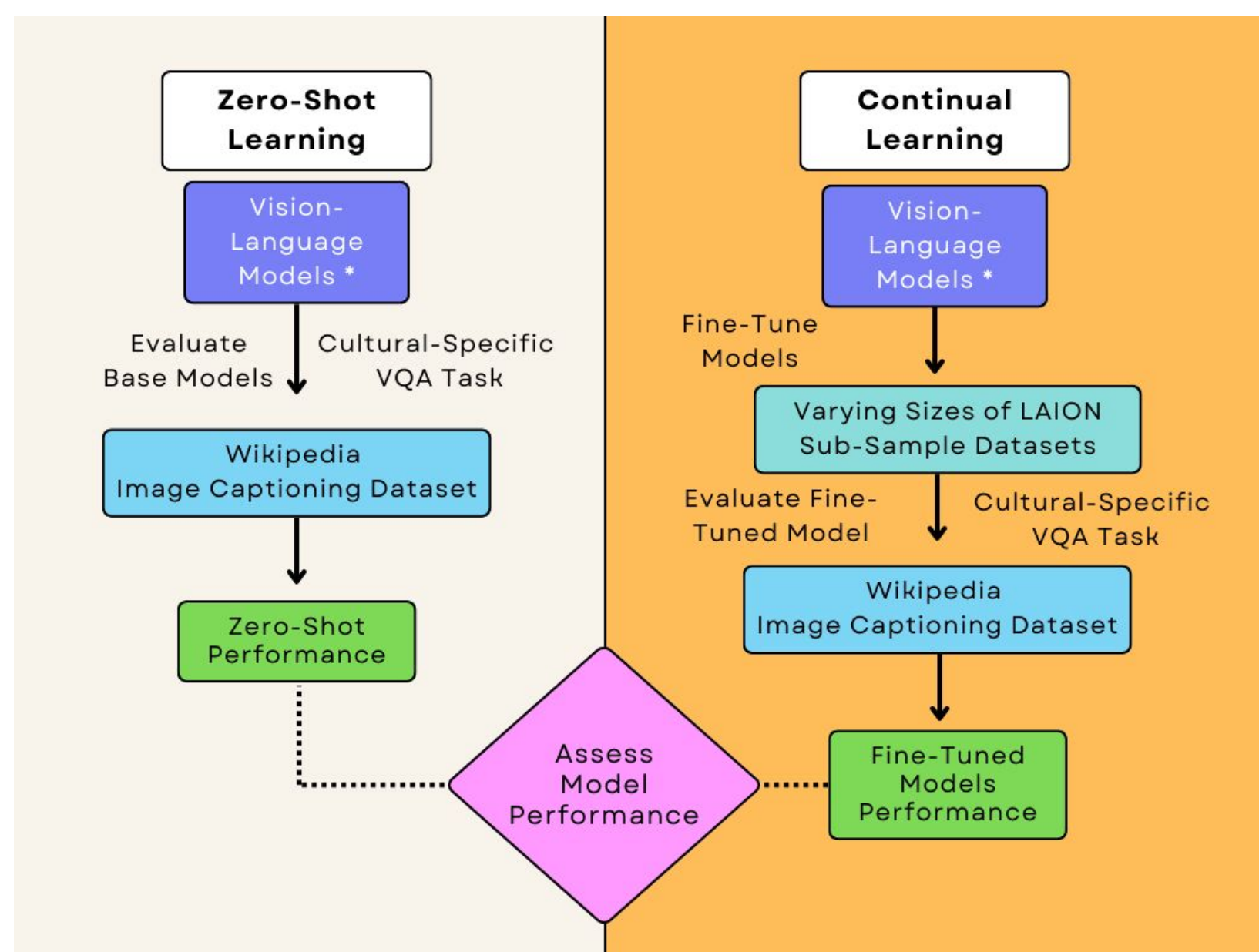
Contributions: Wikipedia Image Caption benchmark + analysis of 3 VLMs with different fine-tuning scales, investigating cultural bias across scale and models.

Methodology

Dataset

LAION - 400M is an open source image-text-pair dataset extracted from the Common Crawl web data dump. The complete dataset contains 50MB of parquet files which was parsed using the img2dataset tool to download and store all the image and text pairs onto HPC.

Experimental Design

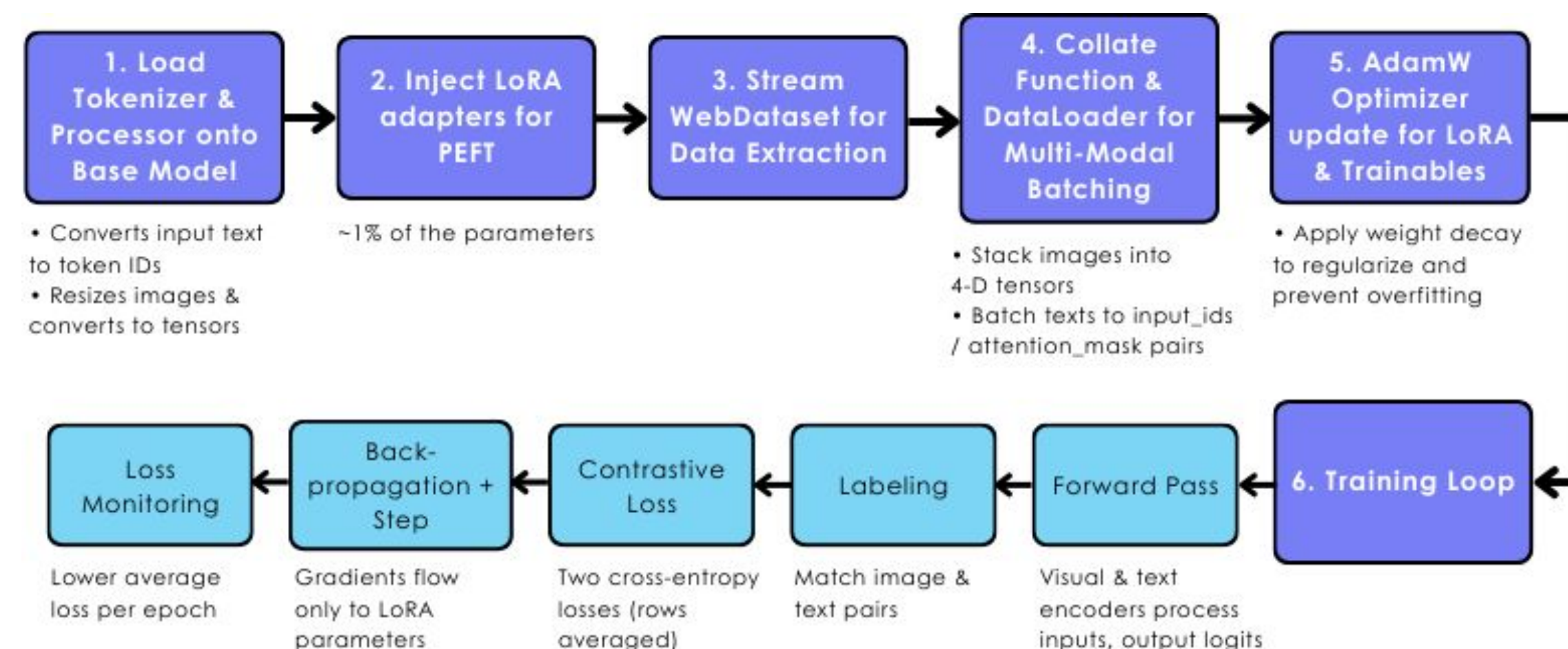


Zero-shot and fine-tuning approach for evaluating models on cultural recognition accuracy

Models

LLaVA-7B, **OpenCLIP-88M**, and **SmoIVLM-2B** — three open-source vision language models (VLM) under 8B parameters — were the selected for evaluation and fine-tuning. LLaVA and OpenCLIP are known for strong multimodal performance, while SmoIVLM is optimized for memory efficiency. Their smaller sizes help isolate the impact of fine-tuning data volume from pre-training effects.

Methodology - Fine-tuning



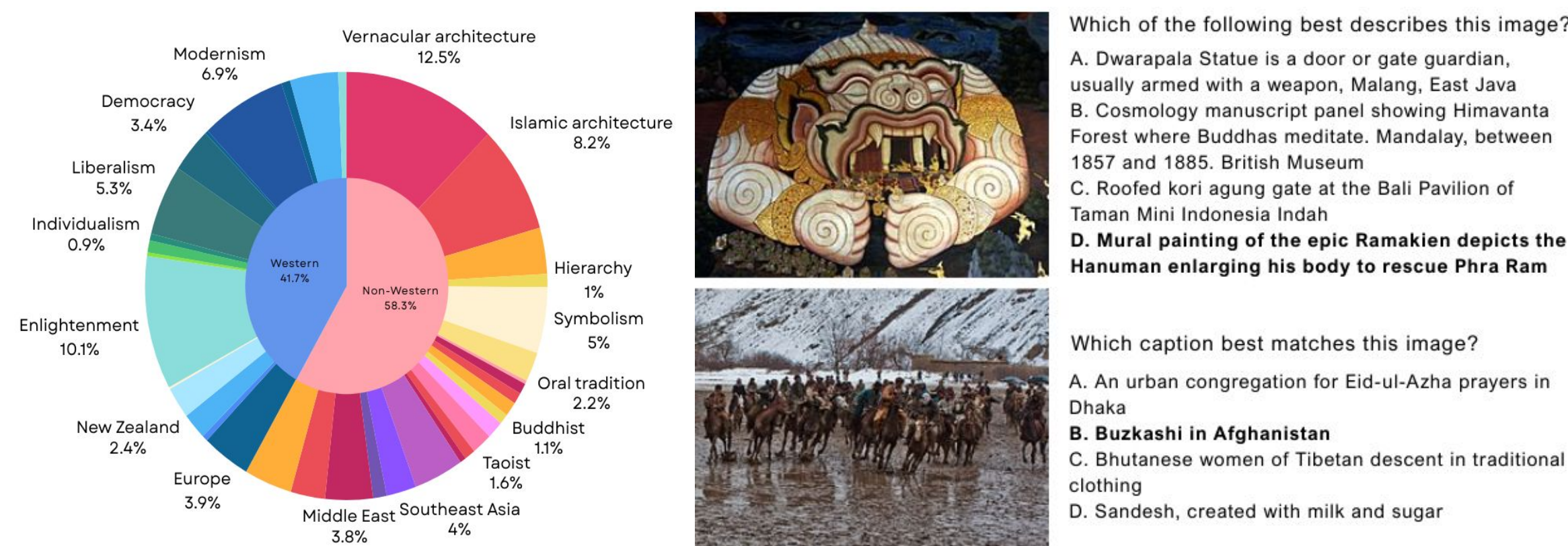
Wikipedia Image Caption Benchmark

Benchmark Construction

We created a lightweight **image caption** evaluation dataset from Wikipedia (n=6408), using a web scraping pipeline. Categories were curated to represent **Western** vs. **Non-Western** cultures, and **continental** groupings. The dataset was stored in a JSON file, containing image URLs, their captions, and their corresponding categories.

Model Evaluation Using Wikipedia Benchmark

Each evaluation was presented with **an image** and **4 multiple-choice captions** — one correct ("ground truth") and three distractors:



Given the image and question, each model was prompted to select the most appropriate answer for each question. Group-specific accuracy was calculated to assess model performance across varying cultural categories.

References

- [1] Amith Ananthram, Elias Stengel-Eskin, Mohit Bansal, and Kathleen McKeown. 2025. See it from my perspective: How language affects cultural bias in image understanding.
- [2] Shudong Liu, Yiqiao Jin, Cheng Li, Derek F. Wong, Qingsong Wen, Lichao Sun, Haipeng Chen, Xing Xie, and Jindong Wang. 2025. CultureVLM: Characterizing and improving cultural understanding of vision-language models for over 100 countries.
- [3] Abeba Birhane, Vinay Prabhhu, Sang Han, and Vishnu Naresh Boddeti. 2023. On hate scaling laws for data-swaps.

Results

- OpenCLIP performs best
- All models have higher accuracy in Western groups compared to Non-Western groups
- Highest accuracy disparities are in LLaVA and SmoIVLM



ECE measures how well the model's confidence matches the actual accuracy by bucketing predictions into confidence intervals.

- North America has highest ECE
- Western ECE is higher than Non-Western ECE

- Z-test: Western vs Non-Western
- Chi Squared test: Continents

Comparison	Test Statistic	P-value	Significant?
SmoIVLM: Western vs Non-Western	0.6560	0.5118	No
SmoIVLM: Continents	11.9749	0.0625	No
LLaVA: Western vs Non-Western	3.3366	0.0008	Yes
LLaVA: Continents	15.0409	0.0199	Yes
OpenCLIP: Western vs Non-Western	2.3341	0.0196	Yes
OpenCLIP: Continents	7.3994	0.2855	No

Conclusion

In conclusion, fine-tuning OpenCLIP with 0.05% and 0.1% of the LAION-400M dataset had no impact on cultural bias. This is likely due to: **1) data leakage**, **2) the format mismatch between fine-tuning and evaluation datasets**, and **3) the insufficient size of the fine-tuning subsets**.

Zero-shot evaluation revealed that:

- LLaVA and OpenCLIP show consistent bias towards Western groups.
- OpenCLIP achieves the highest performance overall.
- SmoIVLM perform more consistently across cultural groups, suggesting stronger cultural inclusivity among the three.

Future Directions

- **Scale fine-tuning** with variety of parameters and **more data** with **efficient hardware** (e.g.: A100 GPUs)
- **Generate confidence scores** for answer choices to better understand models' reasoning
- **Fine-tune on culturally specific dataset** and design a custom evaluation set to avoid data leakage
- **Train VLMs from scratch** to fully isolate & capture the impact of training data composition & size on cultural bias without confounding effects of prior exposure